Stability windows of the Golden Pancake three-body isola: Krein collisions, a half-period reflection symmetry and outer-mass sensitivity

Yuanqing Liu Han Liu Anqing Xuan

Draft, 12 October 2026

Abstract

The *Golden Pancake Drama* orbits are planar, collisionless, zero-angular-momentum periodic solutions of the Newtonian three-body problem with masses $(1, \mu, 1)$, recently reported by the Three Body Orbits atlas on a closed isola $0.4689 \lesssim \mu \lesssim 0.6346$; the atlas asserts linear stability for one representative ($\mu \simeq 0.61236$). We map the planar linear (spectral) stability along the isola. On the segment containing the stable representative we find three spectrally stable stretches, bounded by four transversal Krein collisions of opposite-signature multipliers and by the fold at $\mu_{\max}$. The edges of the main window are $\mu_L = 0.599043784208899851220728860 46\ldots$ and $\mu_R = 0.617844268209063178727366710 60\ldots$ (Table 1); two independent multiprecision pipelines (MPFR Taylor and Gragg–Bulirsch–Stoer integrators with separate shooting and Floquet codes) agree to about 60 significant digits. Near both edges the colliding multipliers split as $|\mu - \mu_*|^{1/2}$ and the monodromy has a 2×2 Jordan block at the collision, the linear signature of a Hamiltonian–Hopf bifurcation. Every orbit on the isola satisfies a half-period reflection symmetry, so the monodromy is a square; this explains why multipliers touch -1 without crossing it (at 1:4 resonances of the half-period map), with no change of spectral stability; possible nonlinear bifurcations there are not investigated. For outer masses $(1 + \delta, \mu, 1 - \delta)$ the orbits remain exactly periodic and the main window closes at $|\delta| \approx 0.0086$. Our orbit at the atlas's 42-digit mass agrees with its published initial conditions to 10^{-36}–10^{-38}. All statements are numerical evidence, not computer-assisted proofs.

1 Introduction

Large numerical searches have produced thousands of planar periodic three-body orbits with zero angular momentum, including many unequal-mass families [3, 4, 5, 6, 7, 8]. Whether a periodic orbit is linearly stable, and how stability changes along a family as the masses vary, has been studied for the figure-eight [9, 10, 11] and, statistically, across the published catalogues [12, 13]. The Three Body Orbits atlas recently reported the *Golden Pancake Drama* family, an isola of orbits with masses $(1, \mu, 1)$ in the Euler-half-twist sector, and found one linearly stable representative at $\mu = 0.612356088827994835\ldots$; its draft note states that "only the marked orbit's linear stability is asserted" [1, 2]. The present note locates and classifies the stability transitions along that isola. The family is not new; its stability structure is the subject here.

2 Model and numerical methods

We integrate $\ddot{\mathbf{q}}_i = \sum_{j \neq i} m_j(\mathbf{q}_j - \mathbf{q}_i)/\|\mathbf{q}_j - \mathbf{q}_i\|^3$, $G = 1$, $(m_1, m_2, m_3) = (1, \mu, 1)$, from the Euler section $\mathbf{q}_{1,2,3} = (-1, 0), (0, 0), (1, 0)$, $\dot{\mathbf{q}}_1 = \dot{\mathbf{q}}_3 = (u, v)$, $\dot{\mathbf{q}}_2 = -2(u, v)/\mu$, which fixes the centre of mass, zero momentum and $L = 0$ [1]. Periodic orbits solve $X(T) - X(0) = 0$ for all twelve labelled positions and velocities by least-squares Newton in (u, v, T); no rotation or body permutation is allowed.

Two independent multiprecision pipelines. (i) An adaptive Taylor integrator in MPFR with Decimal shooting and a symplectic-quotient Floquet reduction; (ii) a Gragg–Bulirsch–Stoer extrapolation integrator in MPFR (modified midpoint rule, step sequence $2j$, Aitken–Neville extrapolation in h^2), mpmath shooting from binary64 seeds, and a reduction-free Floquet analysis. The two share no integration, shooting or spectral code. In (ii), because the eight trivial multipliers of the exact monodromy Φ equal 1, the nontrivial quartic follows from $p_k = \operatorname{tr} \Phi^k - 8$ ($k \leq 4$) by Newton's identities; with pair sums $s = \lambda + \lambda^{-1}$ solving $s^2 - p_1 s + (e_2 - 2) = 0$, the

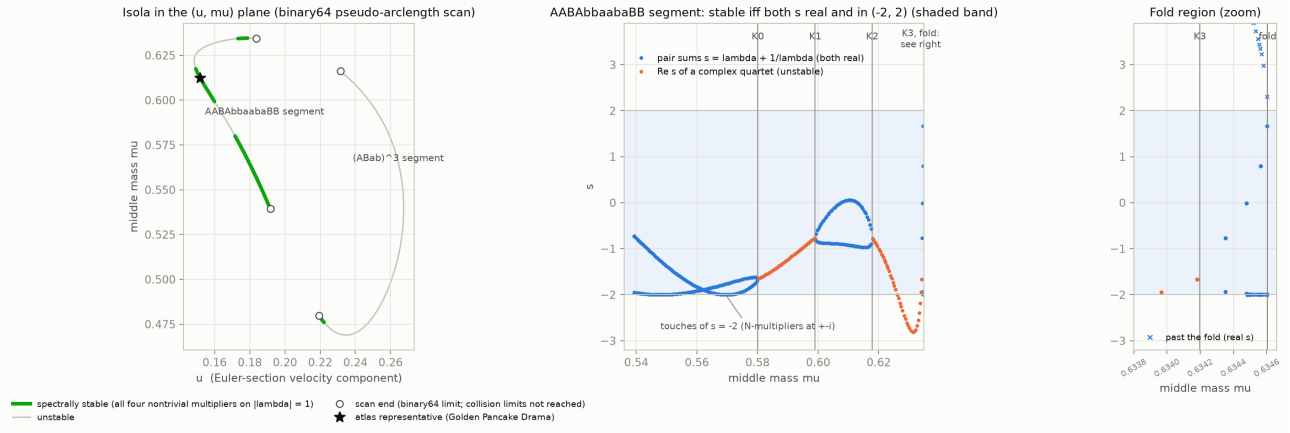


Figure 1: Left: the isola (binary64 pseudo-arclength scan); green, spectrally stable; open circles, ends of the scans before the collision limits; star, atlas representative. Centre: pair sums $s = \lambda + \lambda^{-1}$ on the AABAbbaabaBB segment; blue, both real; orange, real part of a complex quartet. Right: the fold region.

Table 1: Spectral transitions. Digits are truncated where the two independent implementations (Taylor and GBS) agree, excluding each run's last printed digit; all listed transitions have verified sign changes of the transition function at $\mu_* \mp 10^{-30}$.

	type	μ_*	$\mathrm{d}D/\mathrm{d}\mu$
K_0	Krein	0.58009441428846899654889718096487964027 4 . . .	−22.25
$K_1 = \mu_L$	Krein, Jordan	0.59904378420889985122072886046206841927541372566178042574434 7 . . .	+74.87
$K_2 = \mu_R$	Krein, Jordan	0.61784426820906317872736671060041799670207268916883745867616 9 . . .	−305.9
K_3	Krein	0.63419706693055458855260542409815000464 3 . . .	+5840
fold	$s = +2$	≈ 0.6346043 (binary64)	—
T_1	$s = +2$, $(ABab)^3$	0.47590144455086411875206998642346008789 6 . . .	—

discriminant is $D = 2p_2 + 8 - p_1^2 = (s_1 - s_2)^2$. Spectral stability holds iff both s_i are real and in $(-2, 2)$; $D < 0$ signals a complex quartet. Krein forms $\kappa = \mathrm{Re}(i\, z^{\mathsf{H}} \Omega z)$ use full-space eigenvectors and $\Omega = \left(\begin{smallmatrix} 0 & M \\ -M & 0 \end{smallmatrix}\right)$, $M = \mathrm{diag}(1, 1, \mu, \mu, 1, 1)$. Typical settings: 80–90 digits, final closures 10^{-62}–10^{-65}, verification runs with stricter tolerances than the Newton runs. Transitions are refined by secant iteration on D (or $q(\pm 1) = \prod_i (\pm 1 - \lambda_i)$ for $s = \pm 2$ crossings), each evaluation being a fresh Newton-corrected orbit; the sign of the transition function is then verified at $\mu_* \mp 10^{-30}$. Binary64 pseudo-arclength continuation in (u, v, T, μ), with an exact mass-sensitivity column, maps the whole isola.

Validation. On common orbits the pipelines agree to $|\Delta D| \approx 5 \times 10^{-59}$ and $|\Delta \Phi| \approx 10^{-60}$. Continuing our own branch to the atlas's 42-digit mass gives (u, v, T) differing from the atlas's Table 1 [1] by 3.3×10^{-38}, 1.9×10^{-38} and 1.5×10^{-36} (the atlas states a closure of 1.5×10^{-36}), and multipliers agreeing with its 19-digit values to 4×10^{-20}. The largest nontrivial multiplier moduli of the two other atlas orbits on this branch ($\mu = 0.58764$, 0.63031) are 1.16366 and 2.38147, reproducing the catalogue values 1.1637 and 2.3815 [2].

3 Results

Stability along the isola. Figure 1 shows the isola in the (u, μ) plane and the pair sums along the AABAbbaabaBB segment, which contains the atlas representative. Three stretches are spectrally stable: from the end of our scan ($\mu \approx 0.5393$, where the binary64 corrector reaches its accuracy limit before the collision limit near $\mu \approx 0.5066$) up to K_0; the main window $[K_1, K_2]$; and a narrow stretch $[K_3, \mu_{\max}]$ next to the fold, where a multiplier pair passes $+1$ ($\mu_{\max} \approx 0.6346043$ from our scan, 0.6346042508 in [1]). The $(ABab)^3$ segment is unstable except for a short stretch beyond its $\mu_{\min}$ fold, starting at an $s = +2$ crossing T_1 at $\mu = 0.475901444\ldots$. Table 1 lists the refined transitions; every crossing is transversal.

Type of the main transitions. At K_1 and K_2 the two colliding multipliers have opposite Krein signatures.

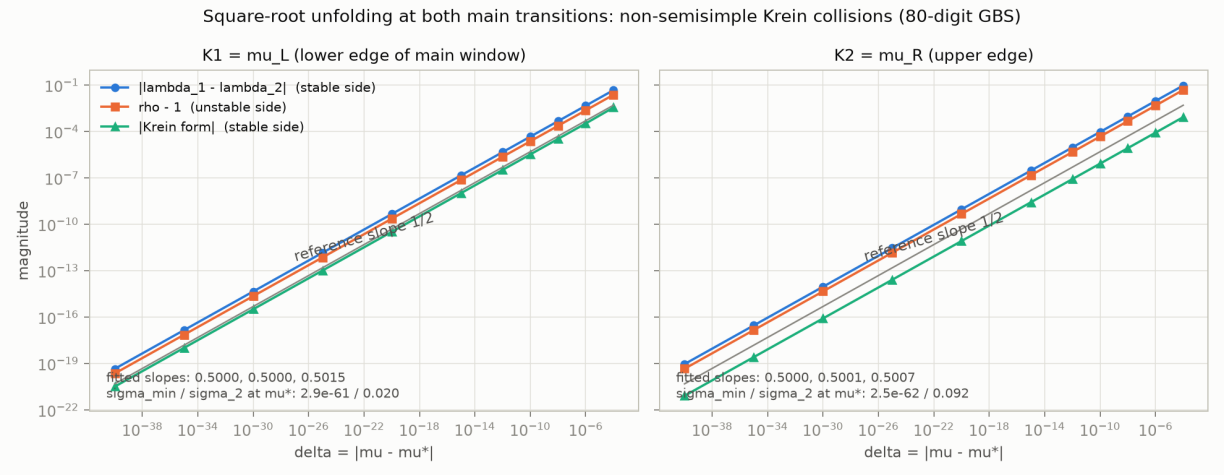


Figure 2: Square-root unfolding of the Krein collisions K_1 and K_2 (80-digit GBS orbits).

Sampling $\mu = \mu_* \pm 10^{-k}$, $4 \le k \le 40$, at 80 digits (Fig. 2) gives on the stable side $|\lambda_1 - \lambda_2| \propto \delta^{0.5000}$ and normalized Krein forms $\propto \delta^{0.501}$, on the unstable side $|\lambda| - 1 \propto \delta^{0.5000}$, and eigenvector non-overlap $\propto \delta^{1.000}$. At μ_* the smallest singular value of $\Phi - \bar{\lambda} I$ is 3×10^{-61} (K_1) and 2×10^{-62} (K_2) while the next is 0.020 and 0.092: the collision eigenvalue has geometric multiplicity one and algebraic multiplicity two. This is the linear part of a Hamiltonian–Hopf bifurcation [14, 15]; we have not computed the nonlinear normal-form coefficient that distinguishes its super- and subcritical cases.

Half-period reflection symmetry. The atlas computes this family in an "Euler-half-twist" symmetric sector with half-period shooting [1]; we make the symmetry and its spectral consequences explicit. For every orbit computed, $X(T/2) = \sigma X(0)$ where σ reflects all positions and velocities across a line through the centre of mass (defect $\le 8 \times 10^{-63}$ at 80 digits; the axis angle varies with μ). Since reflections commute with the Newtonian flow, $X(t + T/2) = \sigma X(t)$ and $\Phi = N^2$ with $N = \sigma\, \mathrm{D}\varphi_{T/2}(X(0))$; numerically $|N^2 - \Phi| \le 3.3 \times 10^{-59}$. Hence $s_\Phi = s_N^2 - 2 \ge -2$ for real s_N: a pair of multipliers of Φ can reach -1 only tangentially, when a pair of N passes $\pm i$, i.e. at a 1:4 resonance of N. The scan shows exactly this near $\mu \approx 0.5695$ and 0.5481; the multipliers of Φ stay on the unit circle, so spectral stability does not change there. The statement is spectral only: 1:4 resonances generically create period-4 orbits of the resonant map, including non-symmetric ones born in pitchfork bifurcations [16], which here would be period-$2T$ orbits of the flow breaking the half-period symmetry; we have not investigated such nonlinear bifurcations. A further near-coincidence near $\mu \approx 0.5619$ is a pass-through of pairs of equal Krein sign, again without a spectral change. The discriminant of N changes sign at K_1 and K_2 together with that of Φ, so these are collisions of the half-period map itself.

Unequal outer masses. For masses $(1 + \delta, \mu, 1 - \delta)$ we replace the symmetric ansatz by a general collinear section with zero momentum and $L = 0$ and solve $X(T) = R_\theta X(0)$ with the rotation angle θ as an unknown. The solutions remain exactly periodic ($|\theta| \le 4 \times 10^{-11}$ in binary64), as the persistence of the reflection symmetry for arbitrary masses suggests. The main window narrows quadratically, $\mu_L(\delta) \approx \mu_L + 126\,\delta^2$, $\mu_R(\delta) \approx \mu_R - 39\,\delta^2$, and closes at $|\delta| \approx 0.0086$ near $\mu \approx 0.611$ (Fig. 3). The scan is binary64 and symmetric under $\delta \to -\delta$ to 3×10^{-11}.

4 Discussion and limitations

The main stability window of the Golden Pancake isola is bounded by Hamiltonian–Hopf-type Krein collisions, and stable motion survives on two further stretches of the same segment. The window is narrow in the outer-mass asymmetry ($|\delta| \lesssim 0.009$), wider than the 10^{-5}-scale stability region reported for the figure-eight [6] but far narrower in δ than in μ. The evidence is numerical: closures and spectra are confirmed by independent integrators, precisions and the atlas's own data, but nothing here is an interval-arithmetic proof of existence or of the spectral statements (compare [11]); local error control is heuristic; the binary64 scans can miss features finer than their step and stop before the binary-collision limits; and only spectral stability is addressed — nonlinear phenomena, such as orbits bifurcating at the 1:4 resonances of the half-period map or the normal-

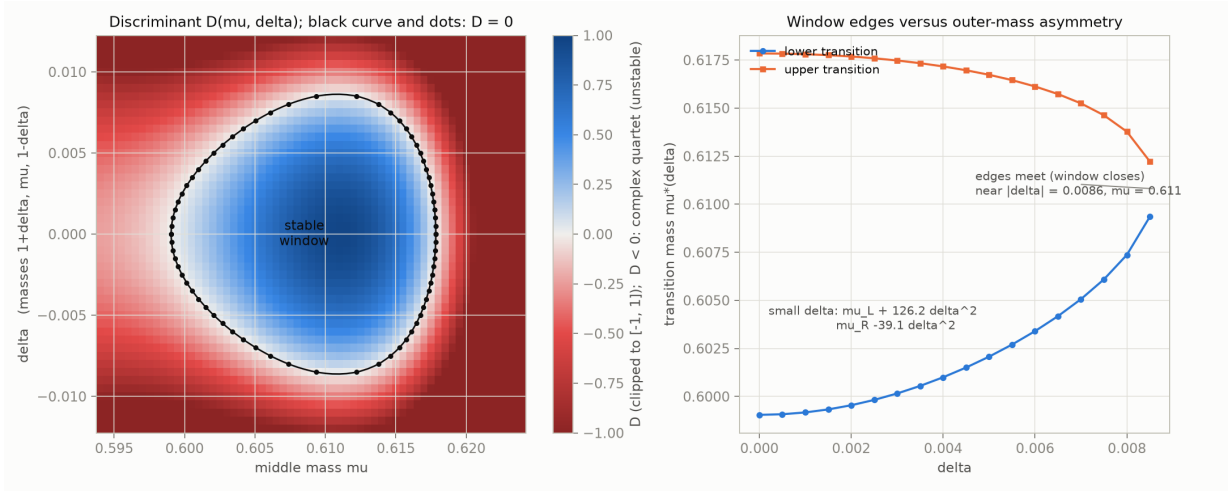


Figure 3: Left: discriminant $D(\mu, \delta)$ for masses $(1+\delta, \mu, 1-\delta)$ (binary64); the black curve is $D = 0$ and bounds the stable window. Right: window edges versus δ.

form type of the Hamiltonian–Hopf transitions, are not.

Data and code. Scripts, raw multiprecision trajectories and monodromy matrices, logs, source snapshots and tests accompany this draft (project repository, `README.md`).

References

[1] Three Body Orbits, "golden pancake drama: a linearly stable three-body orbit on an unequal-mass isola," draft numerical research note, 9 September 2026, https://www.threebodyorbits.com/papers/golden-pancake-drama-2026.pdf.

[2] Three Body Orbits atlas, family page and catalogue, https://www.threebodyorbits.com/family/golden-pancake-drama, https://data.threebodyorbits.com/catalogue.json (accessed 12 October 2026).

[3] M. Šuvakov and V. Dmitrašinović, Phys. Rev. Lett. **110**, 114301 (2013).

[4] X. Li and S. Liao, Sci. China Phys. Mech. Astron. **60**, 129511 (2017).

[5] X. Li, Y. Jing and S. Liao, Publ. Astron. Soc. Japan **70**, 64 (2018).

[6] X. Li, X. Li and S. Liao, Sci. China Phys. Mech. Astron. **64**, 219511 (2021).

[7] I. Hristov and R. Hristova, Astron. Comput. **49**, 100880 (2024).

[8] I. Hristov, R. Hristova and K. Tanikawa, New Astron. **125** (2026).

[9] J. Galán, F. J. Muñoz-Almaraz, E. Freire, E. Doedel and A. Vanderbauwhede, Phys. Rev. Lett. **88**, 241101 (2002).

[10] C. Simó, Contemp. Math. **292**, 209 (2002).

[11] T. Kapela and C. Simó, Nonlinearity **20**, 1241 (2007).

[12] V. Dmitrašinović, A. Hudomal, M. Shibayama and A. Sugita, J. Phys. A **51**, 315101 (2018).

[13] V. Vasiljević, B. Raonić and V. Dmitrašinović, New Astron. **100**, 101969 (2023).

[14] J.-C. van der Meer, *The Hamiltonian Hopf Bifurcation*, Lecture Notes in Mathematics 1160 (Springer, 1985).

[15] J. E. Howard and R. S. MacKay, J. Math. Phys. **28**, 1036 (1987).

[16] M. Gonchenko, S. V. Gonchenko, I. Ovsyannikov and A. Vieiro, Chaos **28**, 043123 (2018), arXiv:1711.09447.